Step 2 : Data

Inspectors	Restaurants		
	I	II	III
1	71	55	84
2	65	57	86
3	70	65	77
4	72	69	70
5	76	64	85

Step 3 : Level of significance $\alpha = 5\%$

Step 4 : Test Statistic

For inspectors: F_{0a} (treatments) = $= \frac{MST}{MSE}$

For restaurants: F_{0b} (blocks) $= \frac{MSB}{MSE}$

Step-5 : Calculation of the Test Statistic

Inspectors	Restaurants			Total $x_{i.}$	$x_{i.}^2$
	I	II	III		
1	71	55	84	210	44100
2	65	57	86	208	43264
3	70	65	77	212	44944
4	72	69	70	211	44521
5	76	64	85	225	50625
$x_{.j}$	354	310	402	1066	
$x_{.j}^2$	125316	96100	161604		

Squares

5041	3025	7056
4225	3249	7396
4900	4225	5929
5184	4761	4900
5776	4096	7225
		$\sum\sum x_{ij}^2 = 76988$

Correction Factor: $CF = \frac{G^2}{n} = \frac{(1066)^2}{15} = \frac{1136356}{15} = 75757.07$

12th_Statistics_EM_Unit_3.indd 97 1/28/2020 6:33:46 PM